LAB 5: Sequence Diagram

Smart Waste Collection Management System

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Prepared by: Tanishka Singh - Leader (Product Manager)

1. Objective

To model the time-ordered interaction between the actors and objects of the Smart Waste Collection Management System using UML sequence diagrams, based on the use cases and requirements in SRS v1.0.

2. Sequence diagrams included

Figure	Scenario	SRS reference
Diagram 1 (page 2)	Sensor measures fill-level every 15 minutes, server stores it, checks the threshold and pushes an alert to admin and driver.	4.1, 4.2 (REQ-BM-1 to 3, REQ-AL-1 to 3)
Diagram 2 (page 3)	Admin requests a route, server computes it using the mapping API, admin may adjust it, route is pushed to the driver who collects bins.	4.3, 5.5 (REQ-RO-1 to 3)
Diagram 3 (page 4)	Citizen submits a complaint and gets a tracking ID; the complaint is resolved when the bin is collected and the citizen checks its status.	4.5, 5.5 (REQ-CF-1 to 3)

3. Lifelines (participants)

Participant	Type	Role in the interaction
Sensor Node	Object	Measures distance, calculates fill percentage, retries failed reads, transmits data.
Backend Server	Object	Receives data, checks threshold, raises alerts, computes routes, resolves complaints.
Database	Object	Stores readings, alerts, routes, bins and complaints.
Push Service (FCM)	Object	Delivers alerts and routes to the admin and driver apps.
Mapping API (OSM)	Object	Provides distance data for route optimization.
Mobile App / Web Form	Object	Interface used by the citizen to submit and track complaints.
Administrator, Driver, Citizen	Actor	Human users of the dashboard and mobile app.

4. Notation used

Symbol	Meaning
Dashed vertical line / box on top	Lifeline of a participant (green = system object, blue = actor)
Thin rectangle on a lifeline	Activation bar - the participant is executing
Solid arrow with filled head	Synchronous message (call)
Dashed arrow with open head	Return message
Arrow looping back to the same lifeline	Self message
loop / alt / opt frames	Repetition, alternatives (if / else) and optional behaviour

Sequence Diagram 1 - Bin Fill-Level Monitoring and Threshold Alert

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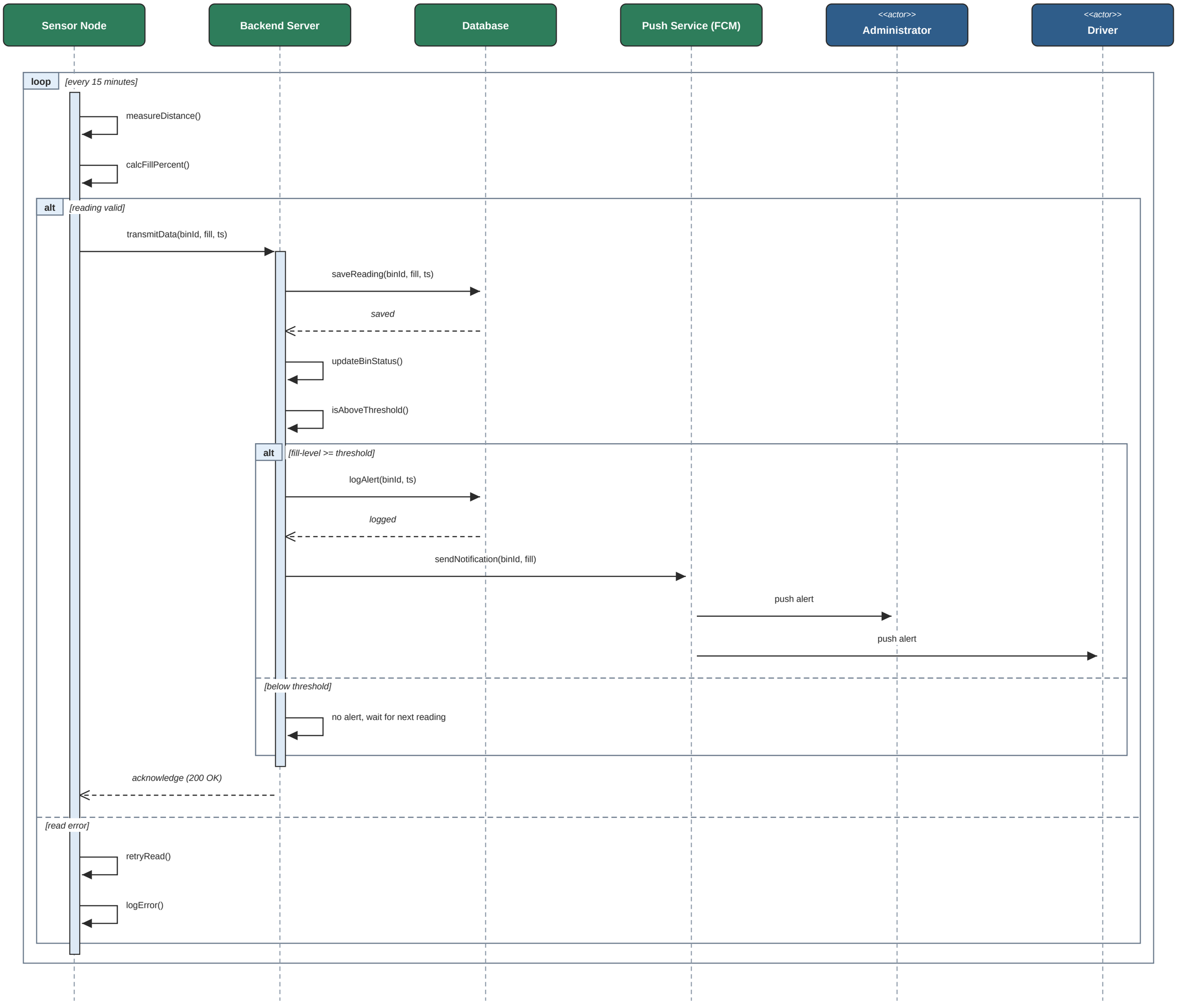

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Sequence Diagram 2 - Route Optimization and Bin Collection

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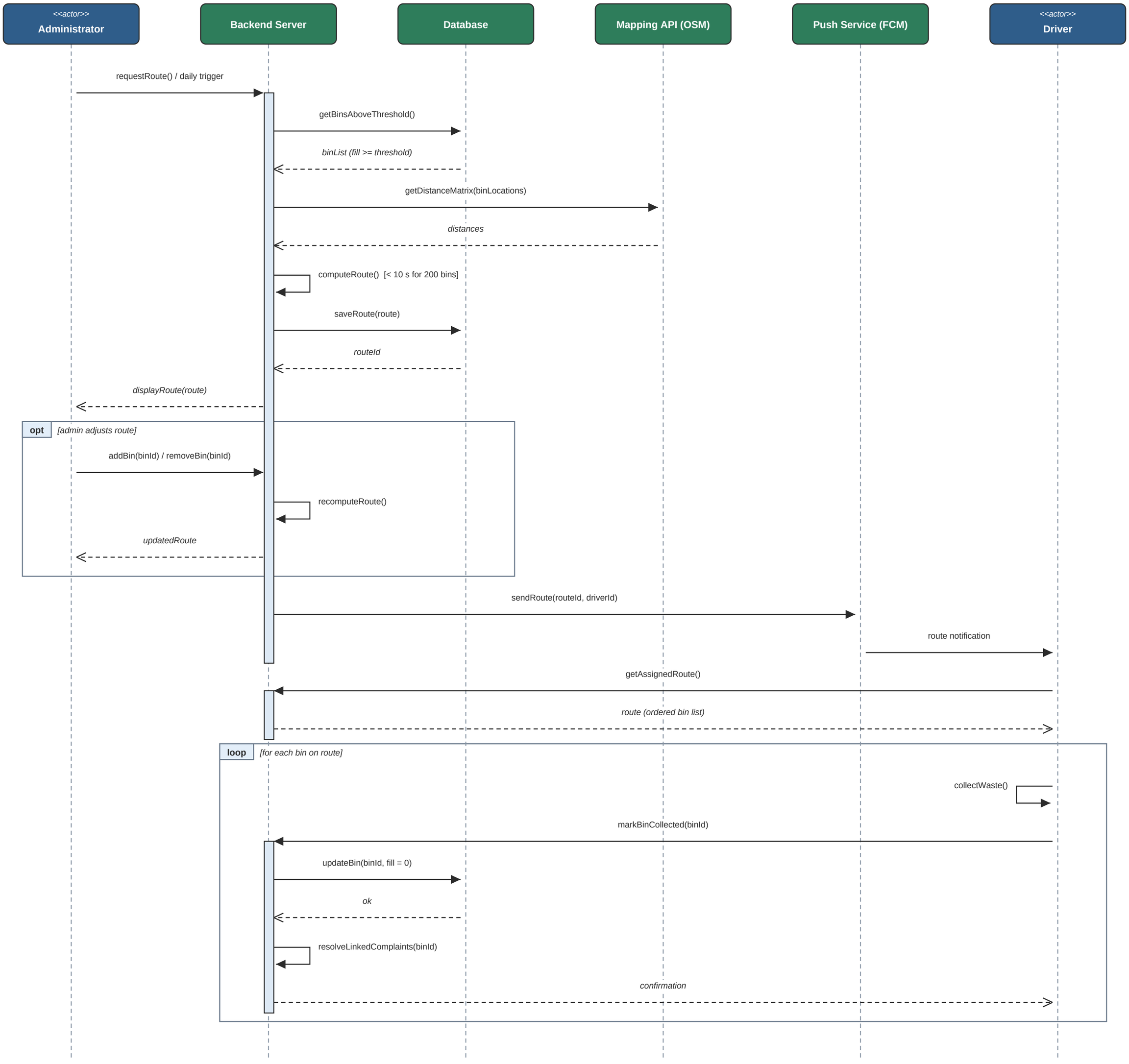

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Sequence Diagram 3 - Citizen Complaint Handling

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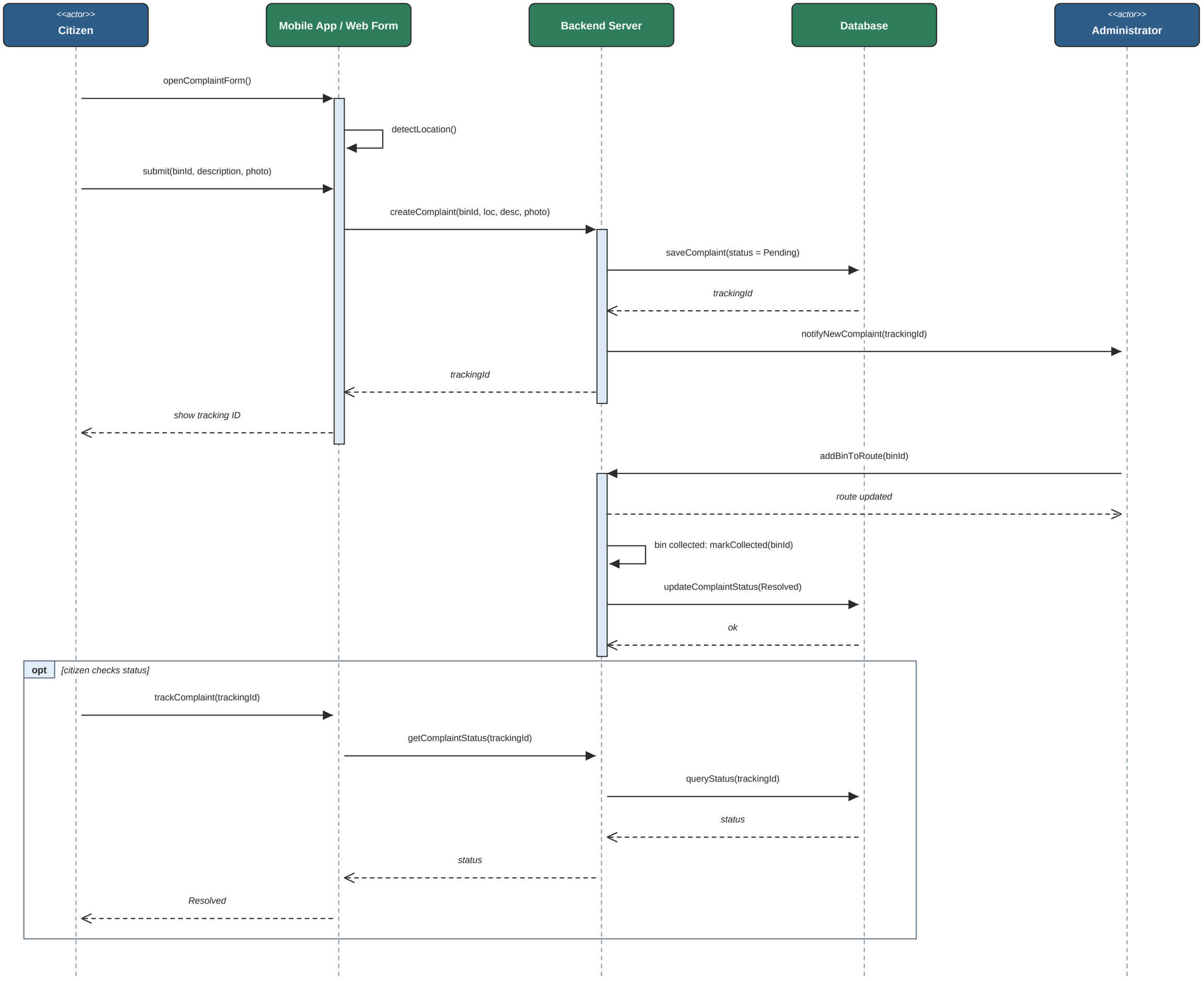

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